



KCB – THIRD-PARTY UTILITY PAYMENTS INTEGRATION DOCUMENT

PREPARED BY	KCB BANK GROUP (PAUL KORE)
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REVIEWER	DAN MUNENE
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Introduction

This API specification document describes the integration process between third-party systems, and the KCB Ecosystem. This document describes in detail the payload structures for the following services and how third parties should conform for this kind of integration.

Pre-requisites

- i. Ensure conformity to the KCB security and authentication standards.
- ii. Ensure the endpoint is http secured(https) with CA signed certificate.
- iii. Ensure conformity to the payloads provided in this document.
- iv. Kindly share the contact details for your developer / technical personnel(over mail) for the integration to ease and fast track the integration time.

API Security

KCB system will sign all data sent from KCB to the 3rd party system and its **REQUIRED** that the 3rd party system verifies the signature sent in the **HTTP** headers using the **PUBLIC KEY** that will be provided by KCB.

This is to ensure that the data has not been tampered with on transit and that it comes from KCB, the issuer of the public key.

NB: This will be verified during UATs to ensure compliance by 3rd party systems.

The expected signature type for verification is **SHA256withRSA**.

The signature field will be sent in the **HTTP headers** of the request with field name **signature** as shown in the below curl request.

```
curl --location --request POST 'http://3rpartyurl.com' \
--header 'signature:
69EJ7+KmfkYHCu7+2mtAks5aFXyQUcEvuZjlpRMEbNszApUymF9eFt25QDb/r0bFP9ZmTNKlhlzdU3cjfBXqs
8Oto9uYcnSsSatxjNhtD21DmAqzYoikJy6zEge2uRaVPzfk9C8eUwlgA5mxLdjWSSib2o8u7WcOTOHpuY8
e6Rk2ydZF3Xuq5e1U9iS6+B+Dw+jqr7a6xTrSHpNQc1BD8FRrKvEK+FtwBArbK9ANT9AD7kQ+AQfJZtljfpzPQ
XnEngOHpt1U1/OstD7I88PrXh9ikeapqqr6/2/t98GxJWABXNjdk8pUMY2SpnsCyoU2BvrNriWOms+YUp3CR
m4g==' \
--header 'Content-Type: application/json' \
```

Services

- i. Bill-validation
- ii. Bill-notification
- iii. Query-transaction

1 BILL-VALIDATION

Transport Protocol	HTTPS
Message Format	JSON
Interaction Type	Synchronous
HTTP Verb	POST
Endpoint	http://xxxxxxxxxxxxxx/xx/xxxxxx

Sample Payload

Sample Request payload

```
{
  "requestId": "d115245e-9604-49de-9436-9fdb539871f",
  "customerReference": "SAMPLE####",
  "organizationReference": "777777"
}
```

Validation Request parameters			
Parameter	Data Type	Mandatory	Description
requestId	String	Yes	Requests identify. Unique for every API call
customerReference	String	Yes	Bill Identify
organizationReference	String	Yes	Organization Identify code at KCB. Issued by KCB

Sample Response Payload

```
{
  "transactionID": "d115245e-9604-49de-9436-9fdb539871f",
  "statusCode": 0,
  "statusMessage": "Success",
  "CustomerName": "Mkenya",
  "billAmount": 100.00,
  "currency": "KES",
  "billType": "FIXED", //FIXED or PARTIAL
  "creditAccountIdentifier": ""
}
```

Validation Response parameters			
transactionID	String	Yes	Transaction identifier. Unique for every API call Return requestID
statusCode	Int	Yes	Code for transaction status 0: Success 1: Failed
statusMessage	String	Yes	Description of the status
CustomerName	String	Yes	Customer name.
billAmount	float	Yes	Transaction amount
currency	String	Yes	Currency field i.e KES
billType	String	Yes	Bill type
creditAccountIdentifier	String	No (Empty string)	Account to be credited (biller account linked to the biller payments to be made)

2 Bill-notification

REQUEST	
Transport Protocol	HTTPS/HTTP
Message Format	JSON
Interaction Type	Synchronous
HTTP Verb	POST
Endpoint	https://XXXXXXXXXXXXXXXXXX/XX/XXXX

Sample Payload

Sample Request payload

```
{
  "transactionReference": "FT00026252",
  "requestId": "c7d702cb-6b5f-4fa6-8b57-436d0f789017",
  "channelCode": "202",
  "timestamp": "20211111030050",
  "transactionAmount": "100.00",
  "currency": "KES",
  "customerReference": "INV-0001",
  "customerName": "John Doe",
  "customerMobileNumber": "25471111111",
  "balance": "100000.00",
  "narration": "Payment for goods",
  "creditAccountIdentifier": "JD001",
  "organizationShortCode": "777777",
  "tillNumber": "150150"
}
```

NOTIFICATION REQUEST PARAMETERS			
Parameter	Data Type	Mandatory	Description
transactionReference	String	Yes	Transaction Identifier. Unique for every transactions.
requestId	String	Yes	Requests identify. Unique for every API call
channelCode	String	No	Code identifying the channel e.g., 202.
timestamp	timestamp(YYYYMMDDHHmmss)	Yes	Transaction timestamp
transactionAmount	String	Yes	Transaction Amount
currency	String	Yes	Transaction currency e.g., KES
customerReference	String	Yes	Unique Identify for every customer
customerName	String	No	The name of the payer.
customerMobileNumber	String	No	The phone number of the payer.
balance	String	No	The amount to settled
narration	String	Yes	The bill narration
creditAccountIdentifier	String	Yes	Account to be credited (biller account linked to the biller payments to be made)
organizationShortCode	String	Yes	unique for every client and the value will be static for every request.
tillNumber	String	No	The collector identifier

Sample Response Payload

NOTIFICATION RESPONSE BODY			
Name	Type	Mandatory	Description
transactionID	String	Yes	Unique for every API call Return RequestID.
statusCode	int	Yes	Code for transaction status 0: Success 1: Failed
statusMessage	String	Yes	Description of the status

```

{
  "transactionID": " c7d702cb-6b5f-4fa6-8b57-436d0f789017",
  "statusCode":0, //0{Success}, 1{Failure}
  "statusMessage": "Notification received" , else error message
}

```

3 Transaction Query API

Version 1.0

Introduction

This document seeks to provide guidance and specifications on how to integrate to the Query Transaction Details API.

Transaction Query API

This is used to retrieve biller transaction records if notifications were not sent. You can fetch using customer reference or the Ft reference.

REQUEST	
Transport Protocol	HTTPS
Message Format	JSON
Interaction Type	Synchronous
HTTP Verb	GET
Endpoint	https://uat.buni.kcbgroup.com/kcb/transaction/query/1.0.0/{INPUT FT OR CUSTOMER REFERENCE}}

Sample Curl

```
curl --location 'https://uat.buni.kcbgroup.com/kcb/transaction/query/1.0.0/184180' \
--header 'accept: */*' \
--header 'Authorization: Bearer eyJ4NXQiOiJNR1F6TmptrelptVTFaV1k0T0' \
--header 'Cookie: 17b6a04d8c33f767f26e4ffbea820ab3=656d7437a7be50266888d8439f792b08;
4b1f380494b4bbde9d5435be5996a54d=656d7437a7be50266888d8439f792b08;
961c85017d5b009458fb3a8ca043506b=668ba184364e7f25359878b72359751e;
d8302f3e4c94982ef82a26c833336c71=a6b3b30720611244e136a93ebc320e2b'
```

Sample Response Payload



```
{
  "creditAccountIdentifier": "1108366384",
  "transactionReference": "123ADP",
  "organizationShortCode": "40019",
  "customerReference": "184171",
  "transactionAmount": "10.0",
  "currency": "KES",
  "statusMessage": "OSP-1002",
  "customerName": "EASY TRADING",
  "statusCode": 0,
  "channelCode": "401",
  "timestamp": "20240418175230"
}
```

Parameter	Data Type	Mandatory	Description
creditAccountIdentifier	String	Yes	Identifier for account
transactionReference	String	Yes	Reference of the transaction
organizationShortCode	String	No	Code identifying the organization e.g., 40019.
customerReference	String	Yes	Unique Identify for every customer
transactionAmount	String	Yes	Transaction Amount
currency	string	Yes	Currency e.g., KES
statusMessage	String	Yes	Status message
customerName	String	Yes	Name of the customer
statusCode	integer	Yes	Status Code e.g., 0
channelCode	String	No	Channel code
timestamp	Timestamp (YYYYMMDDHHmmss)	Yes	Transaction timestamp

